Climate, Fire, and Loss: Predicting Wildfire Insurance Loss in California Using Machine Learning

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Abstract

Wildfire-related insurance losses in California have surged due to severe fire seasons and development in fire-prone areas. Traditional actuarial models use historical insurance metrics, but climate and environmental data may enhance predictions. This study examines if integrating climate, drought, fire perimeter, structure damage data, and modern machine learning improves loss prediction over traditional statistical methods.

Using county level data from the California Department of Insurance, CAL FIRE, NOAA climate, U.S. Drought Monitor, and DINS (2018–2023), I train and evaluate Generalized Linear Model, Random Forest, and XGBoost on two feature sets: an insurance-only baseline and an enhanced set incorporating environmental predictors across 58 California counties.

Enhanced features improve predictive performance, reducing MAE by 6% for GLM, 29% for Random Forest, and 18% for XGBoost. XGBoost leads with a test MAE of $4.8M, outperforming GLM ($12.6M) and Random Forest ($8.8M). SHAP analysis shows insurance-derived features dominate, with drought severity index and temperature as key environmental predictors. GLM coefficient analysis confirms drought index, structure damage, and temperature as significant predictor. Urban counties like Orange, and Riverside show persistent errors due to policy volume, not fire geography in county level analysis.

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1 Introduction

California has faced increasingly destructive wildfire seasons over the past decade, with insured losses rising from hundreds of millions to tens of billions of dollars (California Department of Insurance, 2023). Anthropogenic climate change is a primary driver, significantly increasing wildfire frequency and extent (Williams et al., 2019). The 2018 Camp Fire destroyed more than 18K structures in Butte County, causing extraordinary insurance claims that strained insurers and regulators (California Department of Insurance, 2019). Climate scientists link rising temperatures, prolonged droughts, and increased fire severity to environmental drivers intensifying in the west US (Westerling, 2016; Westerling et al., 2006). Research published, such as Abatzoglou and Williams, 2016, confirms the significant impact of climate change on burned areas expansion in western US forests. Insurance loss prediction traditionally uses actuarial methods focusing on historical claims and policy data. As wildfire risks diverge from historical patterns, the adequacy of these methods is questioned by re-insurers, and the insurance industry.

This Study explores five CA county-level insurance loss prediction questions.. First, it investigates if county-level wildfire insurance losses can be reliably predicted using historical exposure, climate, drought, and fire perimeter data. Second, it compares XGBoost (Chen and Guestrin, 2016) and Random Forest (Breiman, 2001) with Tweedie GLM. Third, it assesses if environmental variables enhance predictions beyond insurance data. Fourth, it examines interpretability by identifying influential factors. Fifth, it evaluates SHAP-based methods (Lundberg and Lee, 2017) for delivering meaningful insights in a regulated insurance context where model transparency is required.

2 Research background

Predicting natural catastrophe losses at a granular geographic level is challenging in actuarial science (Goldburd et al., 2016). The models used by companies like RMS,

AIR Worldwide, and Verisk, etc., use proprietary, computationally expensive stochastic fire spread simulations and exposure databases (Risk Management Solutions, 2022). Statistical GLMs, standard in insurance ratemaking (Goldburd et al., 2016), offer interpretability and regulatory acceptance but struggle with non-linear interactions between environmental and exposure variables.

The insurance loss prediction using machine learning has grown significantly, with gradient boosted trees performance on tabular insurance data as it can model complex features without distributional assumptions (Chen and Guestrin, 2016). Random Forest also performs well in non-linear regression settings where feature interactions and skewed distributions challenge parametric assumptions (Breiman, 2001). However, most ML applications in insurance focus on individual claim severity or frequency rather than aggregate county-level catastrophe losses. While statistical learning methods for actuarial applications have advanced (Wüthrich and Merz, 2023), their use in wildfire catastrophe loss prediction at the county level is underexplored. The environmental research links drought severity, temperatures, and dry conditions to increased wildfire risk and fire season (Westerling, 2016; Westerling et al., 2006). Despite this, their integration into county-level insurance loss prediction models is limited. This paper evaluates the incremental value of climate, drought, fire perimeter, and structural damage data over insurance-only baselines across three models and 58 California counties over six years.

3 Research and Methods

3.1 Problem Formulation

This study addresses wildfire insurance loss prediction as a supervised regression problem, aiming to forecast annual losses at the county level across 58 California counties. For county c in year t, the function to learn is:

$$\hat{y}_{ct} = f(\mathbf{X}_{ct}) \tag{1}$$

where $\hat{y}_{ct}$ is the predicted loss, and $\mathbf{X}_{ct}$ includes insurance, climate, drought, and fire

perimeter features. Two feature sets are tested: an insurance-only baseline and an enhanced set with environmental predictors.

3.2 Experimental Design

A temporal train/test split was used for insurance loss prediction, with 2018–2021 for training and 2022–2023 for testing, avoiding random shuffle to prevent future loss information leakage (Bergmeir and Benítez, 2012). This design mirrors real-world conditions for model generalization to unseen future years across 58 counties.

3.3 Models

3.3.1 Generalized Linear Model

A Tweedie Generalized Linear Model with a log link function was chosen as the actuarial baseline, parameterized by variance power p, where variance is proportional to mean (Nelder and Wedderburn, 1972).

$$V(\mu) = \mu^p \tag{2}$$

A variance power of $p = 1.5$ was selected to accommodate the compound Poisson-Gamma structure of insurance losses, which exhibit both zero values and heavy right tails (Jørgensen, 1987). The log link function ensures non-negative predictions:

$$\log(\mu) = \mathbf{X}\boldsymbol{\beta} \tag{3}$$

where μ is the expected loss, X denotes the features, and β are the weights.

3.3.2 Random Forest

Random Forest is a bagging ensemble averaging predictions across n independently grown decision trees (Breiman, 2001):

$$\hat{y} = \frac{1}{n}\sum_{b=1}^{n} f_b(\mathbf{x})$$

Each tree uses a bootstrap sample and a random subset of features to reduce variance without increasing bias (Breiman, 2001). Key hyperparameters: $n = 200$ trees and *min_samples_leaf* = 2, chosen via grid search to minimize test MAE.

3.3.3 XGBoost

XGBoost is a regularized gradient boosting framework that sequentially fits trees to residuals of prior iterations (Chen and Guestrin, 2016). The objective function minimizes a regularized loss:

$$\mathcal{L} = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \tag{4}$$

where $l(\hat{y}_i, y_i)$ is the prediction loss and $\Omega(f_k) = \gamma T + \frac{1}{2}\lambda\|w\|^2$ penalizes tree complexity through leaf count T and leaf weight magnitude $\|w\|^2$ (Chen and Guestrin, 2016). A regularization parameter of $\lambda = 3$ was applied to reduce overfitting given the limited training set observations. Early stopping with a patience of 30 rounds was used to identify the optimal number of boosting iterations.

3.4 Evaluation Metrics

The 3 metrics are used to assess the model performance, i.e. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the R-squared (R^2) Score . MAE, the primary metric, is directly interpretable in dollar terms and robust to extreme outliers in California wildfire data (Willmott and Matsuura, 2005). RMSE, a secondary metric, penalizes large prediction errors more heavily, crucial for catastrophic years like Butte County 2018. R^2 measures the variance in wildfire losses explained by the model, offering a scale-independent fit measure. However, it was secondary due to its sensitivity to high variance from outlier years, which can inflate model fit without reliable generalization (Willmott and Matsuura, 2005). The formulas are:

$$\text{MAE} = \tfrac{1}{n}\Sigma|y_i - \hat{y}_i|; \quad \text{RMSE} = [\tfrac{1}{n}\Sigma(y_i - \hat{y}_i)^2]^{1/2}; \quad R^2 = 1 - \frac{\Sigma(y_i - \hat{y}_i)^2}{\Sigma(y_i - \bar{y})^2} \tag{5}$$

3.5 Interpretability Approach

Model interpretability was assessed using two complementary approaches reflecting both modern machine learning explainability and traditional statistical inference.

3.5.1 SHAP-Based Explainability

SHAP (SHapley Additive exPlanations) values were calculated for XGBoost on the held-out test set to break down predictions into individual feature contributions. SHAP assigns a contribution value ϕ_i to each feature such that the total of all contributions is equal to the gap between the model's forecast and the global baseline value:

$$f(\mathbf{x}) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{6}$$

where ϕ_0 is the base value representing the mean training prediction and ϕ_i is the marginal contribution of feature i to the individual prediction (Lundberg and Lee, 2017). Two visualizations were created: a side-by-side feature importance bar plot of mean absolute SHAP values for both baseline and enhanced XGBoost models, and a beeswarm plot for the enhanced model only, visualizing each feature's direction and magnitude effect across all test observations, indicating whether high or low feature values push predicted losses upward or downward.

3.5.2 GLM Coefficient Analysis

For the Tweedie GLM, interpretability is inherent, each coefficient β_j directly quantifies the expected change in log predicted loss per unit change in the corresponding standardized feature:

$$\frac{\partial \log(\hat{\mu})}{\partial x_j} = \beta_j \tag{7}$$

Statistical significance was assessed using Wald z-tests with $p < 0.05$ as the threshold, providing a statistically grounded complement to SHAP analysis (Nelder and Wedderburn, 1972).

4 Materials and Data Sources

4.1 Data Sources

Five public data sources were collected for years spanning 2018–2023 across California.

California Department of Insurance (CDI). The primary outcome variable, total wildfire insurance loss, was sourced from the CDI Residential Property Coverage Amounts, Wildfire Risk and Losses dataset (California Department of Insurance, 2023), produced under California Insurance Code Section 929. The dataset includes annual county-level data on exposures, premiums, coverage amounts, and losses for admitted insurers, filtered to retain aggregate loss data meeting the reporting threshold.

CAL FIRE Historic Fire Perimeters. Burned acreage per county per year was derived from CAL FIRE historic fire perimeter (CAL FIRE, 2023).

U.S. Census Bureau TIGER County Boundaries. The U.S. Census Bureau TIGER 2025 (U.S. Census Bureau, 2025) data is used for spatial join with CAL FIRE data using County boundary geometries.

NOAA Climate Data. County-level annual average temperature and precipitation were obtained from NOAA's Climate Data (National Centers for Environmental Information, 2023), aggregating daily station observations to county-year averages.

U.S. Drought Monitor (USDM). Weekly Drought Severity and Coverage Index (DSCI) values were obtained from National Drought Mitigation Center (U.S. Drought Monitor, 2023) and aggregated to annual county-level averages, providing a continuous measure of drought intensity. The DSCI is a composite index that has been widely adopted as a standardized measure of drought severity across the United States (Svoboda et al., 2002).

CAL FIRE Damage Inspection System (DINS). Post-fire structure damage records from the CAL FIRE DINS database (California Department of Forestry and Fire Protection, 2023) were aggregated to county-year level.

4.2 Data Preprocessing

Raw data from five sources were preprocessed into a unified county-year panel dataset. CAL FIRE fire perimeter polygons were joined with county boundaries to calculate annual burned acreage. DINS records were filtered to residential structures and aggregated by county and year. NOAA climate observations and USDM drought values were averaged per county. Insurance records were filtered to exclude non-loss rows, then merged on county name and year into a single dataset.

4.3 Feature Engineering

Derived insurance features. Three features were derived from raw insurance fields: log-transformed exposure, loss per unit exposure, and loss relative to earned premium. These are standard actuarial metrics that normalize losses by exposure and premium volume (Goldburd et al., 2016).

Target encoding. County names were replaced with mean county-level loss computed exclusively from training data, a target encoding approach that converts high-cardinality categorical variables into continuous predictive features (Micci-Barreca, 2001). Applying encoding only on training data prevents data leakage into the test set.

Lag features. One-year lagged versions of all environmental features were constructed to capture the delayed effect of prior-year climate and fire conditions on current-year insurance losses.

Log transformations and standardization. Highly skewed environmental features were log-transformed using $\log(1 + x)$ before GLM fitting to stabilize coefficients. All features were standardized with z-score normalization (Goldburd et al., 2016).

Train/test split. A temporal split was applied with earlier years as training and the most recent two years as the held-out test set.

4.4 Feature Selection

4.4.1 GLM Feature Selection

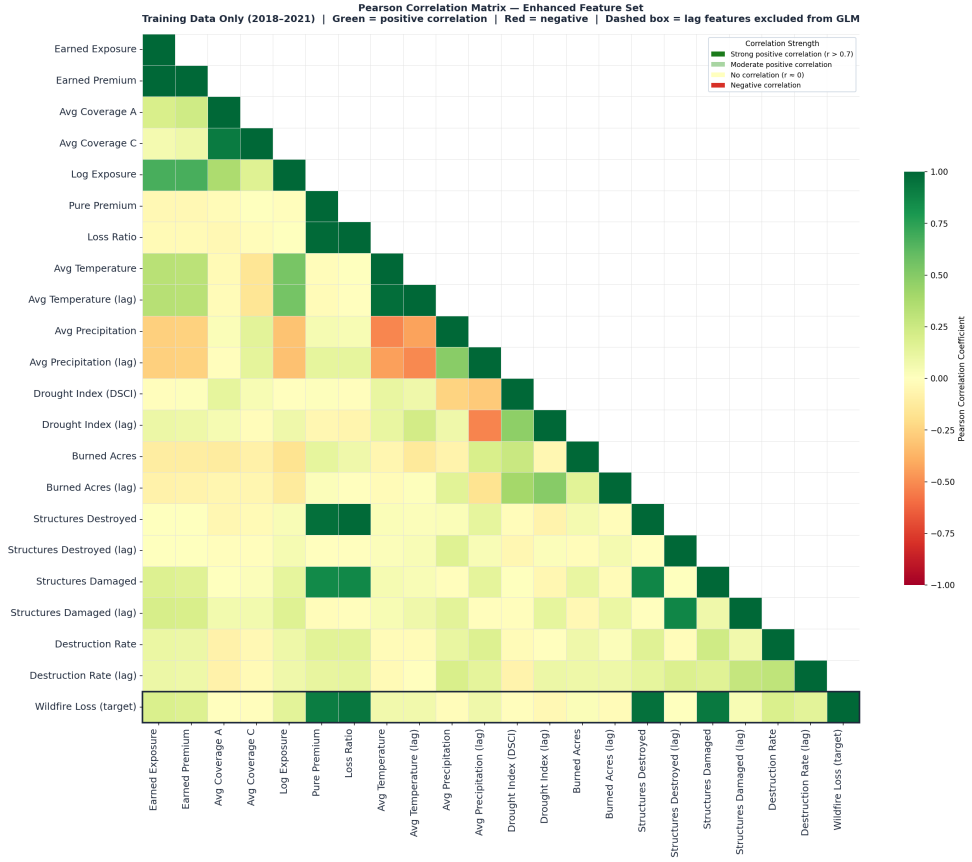


Figure 1

correlation matrix for the enhanced feature set computed on training data only

A Pearson correlation analysis (Schober et al., 2018) identified redundant and multicollinear predictors. Figure 1 shows strong linear relationships between current-year and lagged environmental features such as Avg Temperature, leading to the exclusion of lagged features from the GLM.

Additional, Condition number analysis (Weissfeld and Sereika, 1991) showed severe multicollinearity with Lag features (condition number: 1.48×10^{10} with lags, 27.3 without). Final GLM features: insurance exposure, climate, drought, fire, structure damage.

4.4.2 Random Forest Feature Selection

All features were retained for Random Forest, including lag features and structure damage variables. Tree-based models are insensitive to multicollinearity and feature scale (Breiman, 2001).

4.4.3 XGBoost Feature Selection

The structure damage features were excluded from XGBoost due to performance degradation across various transformations, likely due to the sequential boosting algorithm overfitting to the dominant Camp Fire signal in Butte County 2018 with only four training years per county (Chen and Guestrin, 2016). The final XGBoost feature set comprised insurance, climate, drought, and fire perimeter features including one-year lags.

5 Results

5.1 Insurance-Only Model Performance

Table 1 shows the models performance trained on insurance-only features. XGBoost outperformed both GLM and Random Forest, while all three models improved meaningfully over the naive baseline (global training mean as prediction).

Table 1

Insurance-only model performance. Test set 2022–2023.

Model	MAE	RMSE	R^2
Naive Baseline	\$56.9M	\$66.8M	-0.68
GLM (Tweedie)	\$13.4M	\$32.8M	0.60
Random Forest	\$12.4M	\$44.5M	0.26
XGBoost	\$5.9M	\$13.5M	0.93

5.2 Enhanced Model Performance

Table 2 presents models performance after incorporating climate, drought, fire perimeter, and structure damage features. Each of the three models surpassed their insurance-only

baselines, with Random Forest achieving the greatest relative improvement, and XGBoost maintaining the best overall performance.

Table 2

Enhanced model performance and improvement over insurance-only baseline. Test set 2022–2023.

Model	MAE	RMSE	R^2	MAE Improvement
Naive Baseline	\$56.9M	\$66.8M	-0.68	N/A [1]
GLM (Tweedie)	\$12.6M	\$26.0M	0.74	+5.9%
Random Forest	\$8.8M	\$28.7M	0.69	+29.1%
XGBoost	\$4.8M	\$12.6M	0.94	+19.1%

[1] Naive baseline MAE is feature-independent and unchanged.

6 Discussion

6.1 Predictive Performance

The results in Table 2 and Table 1 confirm that all three models outperform the naive baseline prediction across both feature sets, validating that historical insurance exposure data and feature set combined with environmental data contains sufficient signal for county-level wildfire loss prediction. The XGBoost performance excelled, achieving a best test MAE compare to GLM and Random Forest, is consistent with the established superiority of gradient boosting on tabular insurance data (Chen and Guestrin, 2016). The GLM's competitive R^2 of 0.74 in the enhanced setting despite its linear assumptions confirms that the Tweedie family is well suited to the Poisson-Gamma structure of insurance losses (Jørgensen, 1987), while Random Forest's sensitivity to catastrophic outlier years is reflected in its RMSE. High variance in ensemble methods is a well-documented characteristic when training data contains extreme outliers (Hastie et al., 2009). Figure 2 illustrates that all three enhanced models learned genuine predictive structure, with points clustering along the diagonal and XGBoost showing the tightest alignment.

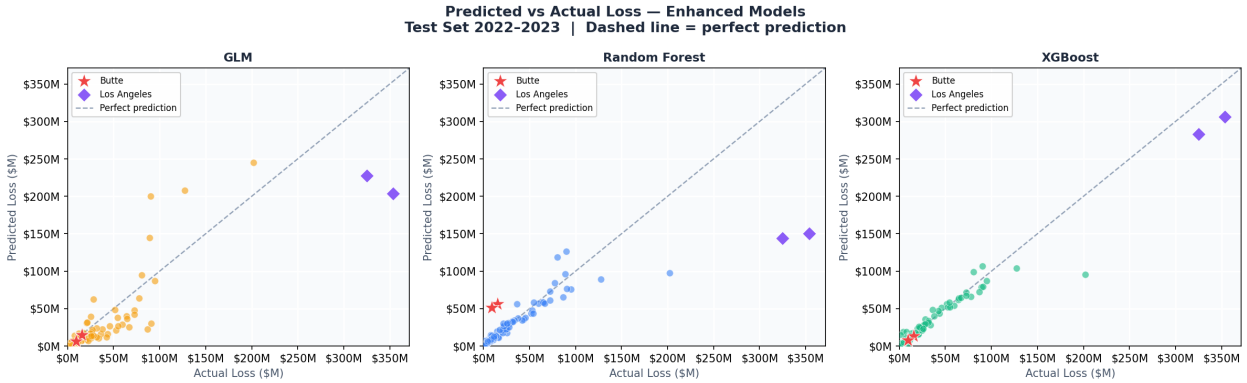


Figure 2

Predicted vs actual losses, enhanced models. Dashed line = perfect prediction.

6.2 Value of Environmental Features

As shown in Table 2, all three models improved with enhanced features, confirming that climate, drought, and fire perimeter data provide incremental predictive value beyond insurance-only baselines. The variation in improvement magnitude across models is instructive. Random Forest showed the largest relative improvement (+29.1% MAE) driven primarily by structure damage features, which tree ensembles can leverage through non-linear interactions without distributional assumptions (Breiman, 2001). XGBoost improved by 19.1% through climate, drought, and fire features. With limited training observations and feature, model selection procedures become critical to avoid overfitting to unrepresentative samples (Arlot and Celisse, 2010). GLM showed the smallest improvement (+5.9%), consistent with linear models' limited capacity to capture non-linear interactions between environmental and exposure variables (Goldburd et al., 2016). Figure 3 illustrates these differential improvements across all three metrics.

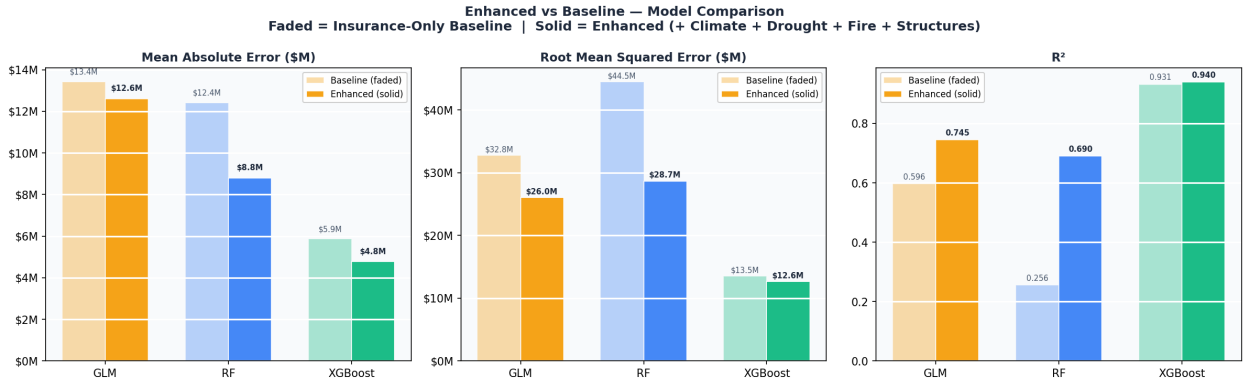


Figure 3

Baseline vs enhanced performance. Faded = insurance-only, solid = enhanced.

6.3 SHAP Feature Importance

Figure 4 compares mean absolute SHAP values between insurance-only and enhanced XGBoost models. Insurance-derived features dominated predictions in both settings, while environmental features contributed meaningful incremental signal in the enhanced model.

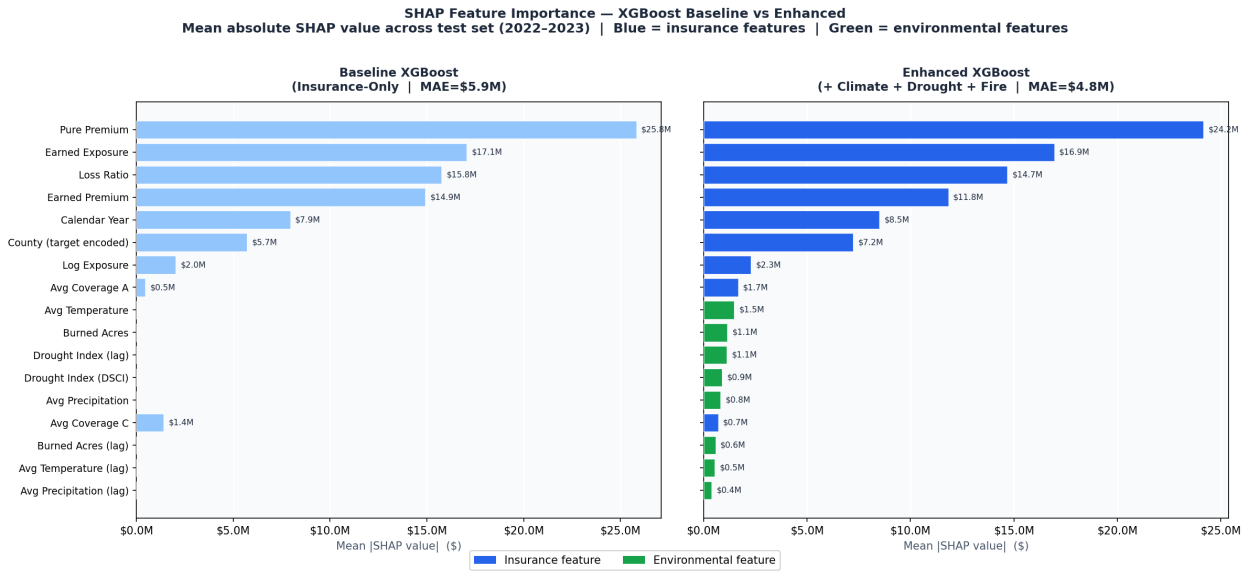


Figure 4

Mean absolute SHAP values, insurance-only vs enhanced XGBoost. Blue = insurance features, green = environmental features.

6.4 Feature Interpretability

Figure 5 presents the SHAP beeswarm plot for the enhanced XGBoost model. Pure Premium and Loss Ratio, both derived from historical losses, functioned as dominant actuarial anchors, consistent with their established role in insurance ratemaking (Goldburd et al., 2016). Among the environmental features, average temperature and drought severity index pushed predicted losses upward, providing model-level confirmation of the established climate-wildfire link (Westerling et al., 2006). The most influential predictors are therefore historical loss-derived features followed by temperature and drought signals.

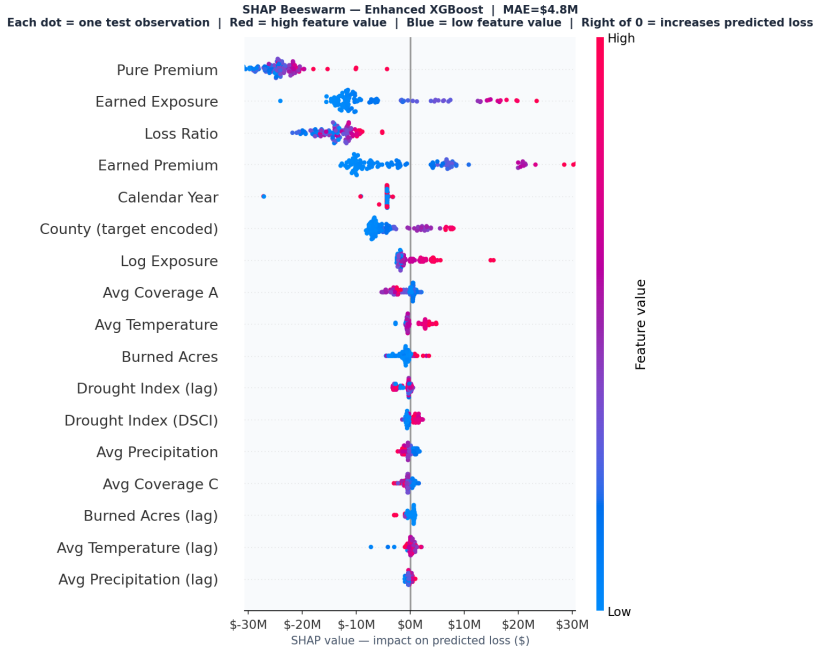


Figure 5

SHAP beeswarm, enhanced XGBoost. Red = high feature value, blue = low. Right of zero = increases predicted loss Lundberg and Lee, 2017.

Table 3 shows GLM coefficient analysis where Drought Index (DSCI), structure damage, and temperature are significant environmental predictors ($p < 0.05$), unlike precipitation and destruction rate. SHAP rankings and GLM significance align, highlighting drought and temperature as key signals, boosting confidence in these findings. SHAP offers directional and distributional insights beyond GLM coefficients, proving SHAP method's value

for actionable insights in insurance, aligning with recent interpretability research (Noll et al., 2020).

Table 3

GLM Tweedie coefficients, environmental features. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

Feature	**Coef**	**Std Err**	**z**	**p-value**
Avg Temperature	-0.175	0.082	-2.13	0.033*
Avg Precipitation	-0.072	0.073	-0.99	0.325
Drought Index (DSCI)	0.245	0.061	3.98	<0.001***
log(Burned Acres)	-0.138	0.077	-1.79	0.073
log(Structures Destroyed)	0.401	0.151	2.66	0.008**
log(Structures Damaged)	0.423	0.123	3.44	0.001***
Destruction Rate	-0.032	0.063	-0.51	0.610

6.5 County-Level Analysis and Limitations

Figure 6 shows XGBoost's geographic consistency, with counties directly exposed to wildfires, benefiting the most from enhanced features. Northern California counties like Siskiyou, Napa, and Plumas, where fire perimeter and climate affect insurance losses, gained significantly from drought, temperature, and burned acreage data. This aligns with documented links between environmental conditions and wildfire activity in California (Westerling, 2016).

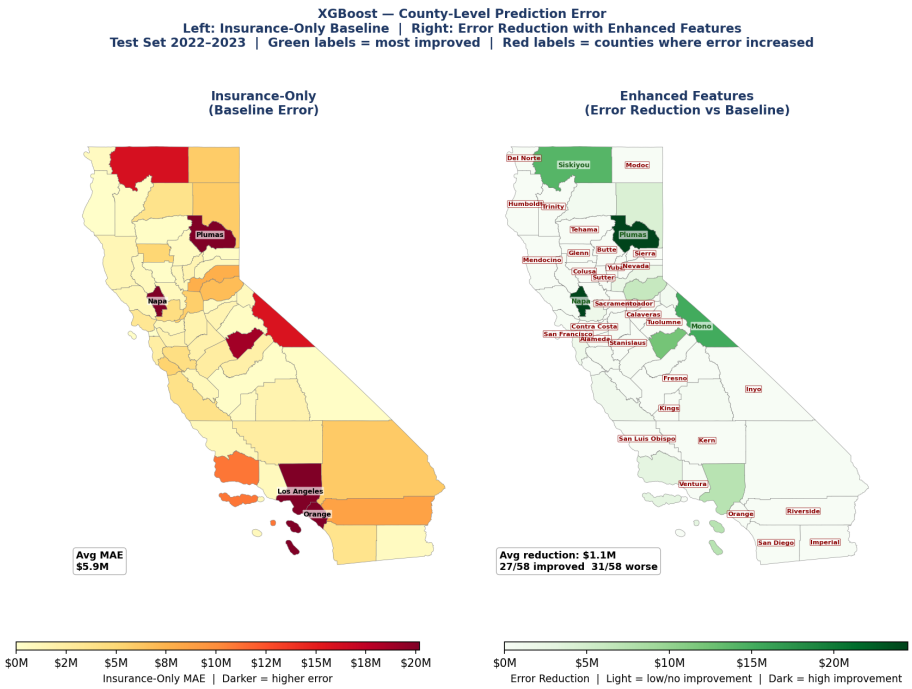


Figure 6

XGBoost county-level prediction error. Left = insurance-only MAE, right = error reduction with enhanced features. Green labels = most improved, red labels = counties where error increased.

Urban southern counties like Orange, and Riverside, showed persistently high prediction error. This reflects a structural limitation rather than a model failure: losses in these densely populated counties are driven by policy volume, liability claims, and smoke-related damage across millions of insured properties, not by fire perimeter geography (American Academy of Actuaries, 2019).

The study's limitation is its four-year training set per county, restricted by the CDI 2018-2023 dataset, affecting model generalization across diverse fire seasons (California Department of Insurance, 2023).

7 Conclusion

This study demonstrates that predicting county-level wildfire insurance losses in California, is more precise when incorporating historical insurance exposure and environmental

data. The Tweedie GLM, Random Forest, and XGBoost models all enhanced their predictive accuracy by including climate, drought, fire perimeter, and structure damage attributes. XGBoost consistently outperformed, confirming gradient boosting's superiority over traditional actuarial methods (Chen and Guestrin, 2016), aligning with the increasing use of machine learning in insurance (Escoto et al., 2021).

SHAP analysis and GLM coefficient estimation identifies drought severity index and average temperature as key predictors, confirming SHAP's value in insurance modeling. The SHAP rankings and significant GLM coefficients alignment validate these environmental signals as genuine features.

County-level analysis shows environmental features add the most value in northern fire-prone counties, where losses are linked to climate and fire conditions, while urban counties are harder to predict due to exposure volume rather than geography (Radeloff et al., 2018). With climate projections indicating rising wildfire growth risk in California (Brown et al., 2023), integrating environmental features into insurance loss models is crucial. Wildfire losses disproportionately burden California municipalities, impacting insurance markets and public fiscal systems (Liao and Kousky, 2022). Future work should explore finer spatial resolution, longer training windows, and property-level exposure indices to improve generalization across California's wildfire risk landscape.

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